

T0 Theory and Coherent Ising Machines: Structural Parallels between Deterministic Geometry and Photonic-Based Combinatorial Optimization

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- Doc. 161, table of Ising mappings: title “NP-hard problems” including RSA factorization → “optimization problems (mostly NP-hard)”; factorization is not known to be NP-hard.

Abstract

We analyze the structural parallels between the T0 time-mass duality theory and the Coherent Ising Machine (CIM), a photonic analog computer for solving combinatorial NP-hard optimization problems. We show that both systems rest on the same fundamental principle: *optimization as physical relaxation into a geometric minimum* – not as algorithmic iteration. Six structural parallels are identified, covering determinism, ground state, qubit structure, coupling matrix, fractal corrections, and toroidal topology. The central consequence is the concept of a *T0 Ising Machine*, in which all couplings J_{ij} are not externally programmed but geometrically emerge from the intrinsic time field $T(x, t)$.

.1 Introduction

Motivation

The Coherent Ising Machine (CIM) is an optical analog computer that solves combinatorial optimization problems by physical relaxation into the ground state of an Ising Hamiltonian [1, 2]. The fundamental principle – the system “falls” physically into its solution – stands in direct conceptual relation to the T0 theory [5], which derives all fundamental constants from a single geometric parameter $\xi = 4/30000$.

This document investigates this connection systematically. We show that T0 not only provides a conceptual parallel to the CIM, but enables a fundamentally stronger formulation: a T0-based Ising Machine in which the coupling matrix J_{ij} geometrically emerges from the mass field $m(x, t)$ – rather than being externally programmed.

Structure

Section .2 describes the Ising model as a universal optimization language. Section .3 explains the operating principle of the CIM. Section .4 summarizes the relevant T0 foundations. Section .5 systematically analyzes the six structural parallels. Section .6 formulates the concept of the T0 Ising Machine. Section .7 discusses open research questions.

.2 The Ising Model as Universal Optimization Language

The Hamiltonian

The Ising model from solid-state physics describes magnetic spins $\sigma_i \in \{-1, +1\}$ interacting through a Hamiltonian:

$$H_{\text{Ising}} = -\frac{1}{2} \sum_{i \neq j} J_{ij} \sigma_i \sigma_j - \sum_i h_i \sigma_i \quad (1)$$

Here J_{ij} are the coupling strengths between spins i and j , and h_i are external fields (bias terms). The ground state – the spin configuration with minimal H_{Ising} – corresponds to the optimal solution of the encoded problem.

QUBO: The Universal Mapping

Every NP-hard combinatorial problem can be formulated as minimization of the Ising Hamiltonian (1) via *Quadratic Unconstrained Binary Optimization* (QUBO):

Table 1: Examples of the mapping of optimization problems (mostly NP-hard) onto Ising Hamiltonians

Optimization Problem	Ising Correspondence
Travelling Salesman Problem	Couplings J_{ij} as distance matrix
MaxCut (graph partitioning)	Spins represent node assignments
Portfolio optimization	Spins as stock selection (buy/not buy)
Protein structure prediction	Spins as torsion angle configurations
RSA factorization	Spins as bit representation of factors

(Corrected on 30 Sept. 2026: previously table title “mapping of NP-hard problems” including RSA factorization – factorization is not known to be NP-hard (it lies in $\text{NP} \cap \text{coNP}$); it can be encoded as a QUBO but does not belong among the NP-hard examples.)

This mapping is not an approximation – it is exact. Finding the ground state of (1) solves the optimization problem.

Central Principle

Combinatorial optimization $\equiv$ search for the ground state of the Ising Hamiltonian. This is the conceptual core of both the CIM and – as we will show – the T0 theory.

.3 Coherent Ising Machine: Operating Principle

Physical Implementation

A Coherent Ising Machine (CIM) physically implements the Ising model through a network of time-multiplexed degenerate optical parametric oscillators (DOPO) [3, 4]:

- **Spin** $\sigma_i = +1$: laser pulse with phase 0°
- **Spin** $\sigma_i = -1$: laser pulse with phase 180°
- **Coupling** J_{ij} : electronic feedback network between pulses
- **Dynamics**: physical relaxation into the energy minimum

The system evolves deterministically according to the field equations of the parametric oscillator – and in doing so automatically finds (or with high probability finds) the ground state of the Hamiltonian (1).

Scaling and State of the Art

Table 2: Comparison: Classical computer vs. Coherent Ising Machine

Property	Classical Computer	Coherent Ising Machine
Parallelism	Sequential / limited	Massively parallel (all spins simultaneous)
Solution paradigm	Algorithm iterates	Physical relaxation
Scaling	Exponentially harder	Natural via light propagation
Energy demand	High for NP-hard problems	Potentially orders of magnitude lower
Spins (as of 2024)	–	Up to 100,000 optical spins

Determinism Behind Apparent Stochasticity

A CIM appears to search stochastically: different runs sometimes yield different solutions. But this is a surface property: physically the system follows *deterministically* the field equations of the OPO. The apparent randomness is solely ignorance of the exact initial conditions of the quantum vacuum.

This is the first and most fundamental parallel to the T0 theory.

.4 T0 Foundations: Relevant Concepts

The Fundamental Binding Condition

The T0 theory postulates time-mass duality as a fundamental geometric constraint:

$$T \cdot m = 1 \quad (\text{in natural units}) \quad (2)$$

The intrinsic time field of a particle with mass m is:

$$T(x, t) = \frac{\hbar}{m(x, t) \cdot c^2} \quad (3)$$

Time and mass are not independent quantities – they are reciprocal aspects of a single geometric reality.

The Geometric Origin Parameter

All fundamental constants emerge from a single dimensionless geometric parameter [6]:

$$\xi = \frac{4}{30000} \approx 1.333 \times 10^{-4} \quad (4)$$

This parameter defines the ratio between the Planck scale and the cosmic scale. There are **no free parameters** – the ground state of the theory is fully determined by ξ .

T0 Qubits

In the T0 theory, qubits emerge naturally from time-mass duality [7]:

$$|\text{T0-Qubit}\rangle = \alpha |T\uparrow, m\downarrow\rangle + \beta |T\downarrow, m\uparrow\rangle \quad (5)$$

with the binding condition $T \cdot m = 1$. The spin states are not chosen arbitrarily – they are forced by fundamental geometry:

- $|T\uparrow, m\downarrow\rangle$: high time / low mass
- $|T\downarrow, m\uparrow\rangle$: low time / high mass

The Fractal Correction

The fractal correction K_{frak} plays a central role in T0: $K_{\text{frak}}^{3/2}$ explains the $\sim 2\%$ deviation in the anomalous magnetic moment of the muon (g-2 anomaly) [8]. This is structurally analogous to the annealing mechanism in the CIM.

.5 Six Structural Parallels: T0 and CIM

Parallel 1: Determinism Behind Apparent Stochasticity

Table 3: Parallel 1: Determinism

Aspect	Coherent Ising Machine	T0 Theory
Surface	Apparently stochastic search	Apparently probabilistic QM
Deep structure	Deterministic (OPO field eqs.)	Deterministic ($m(x, t)$ geometry)
Source of "randomness"	Unknown initial conditions	Unknown field configuration
Consequence	Solution is physically forced	Measurement result geometrically fixed

Both systems replace fundamental probabilism with hidden determinism: in the CIM the solution is not computed, it *emerges*. In T0 the measurement result is not chosen randomly – it is already determined by the mass field $m(x, t)$.

Parallel 1

Apparent probabilism $\equiv$ hidden determinism through geometric field structure – in CIM and T0.

Parallel 2: Ground State as Geometric Minimum

The CIM solves optimization through physical relaxation:

$$H_{\text{Ising}} \rightarrow \text{minimum} \equiv \text{optimization solution} \quad (6)$$

T0 describes the universe as already residing in the ground state of its own geometry:

$$\xi = \frac{4}{30000} \equiv \text{geometric minimum} \Rightarrow \text{all fundamental constants} \quad (7)$$

In both cases the "solution" is not a computation – it is the physically natural resting state of the system. **Optimization is relaxation, not iteration.**

Parallel 3: Qubit Structure

Table 4: Parallel 3: Comparison of qubit structures

Aspect	CIM Spin	T0 Qubit
Carrier	Phase of laser pulse	Time-mass relation
Spin +1	Phase 0°	$ T\uparrow, m\downarrow\rangle$
Spin -1	Phase 180°	$ T\downarrow, m\uparrow\rangle$
Constraint	External (laser pump)	Intrinsic: $T \cdot m = 1$
Superposition	Coherent phase superposition	$\alpha T\uparrow, m\downarrow\rangle + \beta T\downarrow, m\uparrow\rangle$
Collapse	Phase measurement	Geometric field projection

Fundamental Difference

The T0 spin is *more fundamental* than the CIM spin: it carries its own constraint ($T \cdot m = 1$) intrinsically. A CIM spin must be externally constrained to ± 1 ; a T0 spin is constrained by the geometry of spacetime itself.

Parallel 4: The Coupling Matrix J_{ij}

This is the deepest and physically most significant parallel:

Table 5: Parallel 4: Coupling matrix – the decisive difference

Aspect	Classical CIM	T0 Ising Machine (concept)
Couplings J_{ij}	Externally programmed	Geometrically emergent
Degrees of freedom	J_{ij} freely selectable	Uniquely fixed by field geometry
Physical meaning	Mathematical auxiliary	Real geometric interactions
Universality	Arbitrary problems encodable	Physically meaningful problems
Advantage	Maximum flexibility	No encoding errors possible

In a T0-based Ising Machine, J_{ij} is *not* a free parameter – it emerges directly from the intrinsic time field:

$$J_{ij}^{(T0)} = \int T(x, t)_i \cdot T(x, t)_j \cdot G(x_i, x_j) d^3x \quad (8)$$

where $G(x_i, x_j)$ is the geometric propagator of the T0 mass field.

Parallel 5: Fractal Correction as Annealing Mechanism

Table 6: Parallel 5: Fractal correction and annealing

Aspect	Simulated Annealing (CIM)	T0 Fractal Correction
Parameter	Temperature T_{ann}	Correction factor K_{frak}
Function	Prevents local minima	Explains systematic deviations
Effect	Cooling $\rightarrow$ ground state	Scaling $\rightarrow$ precision
Example	MaxCut optimization	g-2 anomaly: $K_{\text{frak}}^{3/2}$ explains $\sim 2\%$
Mechanism	Thermal fluctuations	Fractal self-similarity

The fractal correction $K_{\text{frak}}^{3/2}$ in T0 plays the analogous role of the annealing temperature parameter: it prevents the system from becoming trapped in a local minimum of the geometric energy landscape.

Parallel 6: Toroidal Topology as Optimal CIM Geometry

T0 describes the universe as a 4D torus crystal [9]. This topology has direct implications for a T0-based Ising Machine:

- **Toroidal boundary conditions:** no boundary problem in optimization – all spins have equally many neighbors
- **Periodic structure:** natural implementation of translation-invariant couplings

- **4D instead of 3D:** richer coupling structure than classical 2D/3D CIM arrays
- **Crystal symmetry:** discrete symmetry groups → natural spin lattices without external structure specification

Parallel 6

A T0 Ising Machine would operate on the *natural geometry of the universe* – not on an artificially constructed graph.

.6 The T0 Ising Machine: Conceptual Formulation

Overall Structure

Based on the six identified parallels, a T0-based Ising Machine can be conceptually defined:

Table 7: Overall comparison: Classical CIM vs. T0 Ising Machine

Component	Classical CIM	T0 Ising Machine
Spin carrier	DOPO laser pulses (phase)	Time field states $T(x, t)$
Spin values	± 1 (phase $0^\circ/180^\circ$)	$ T\uparrow, m\downarrow\rangle, T\downarrow, m\uparrow\rangle$
Couplings	Externally programmed	Geometrically emergent from $m(x, t)$
Topology	Arbitrary graph	4D torus crystal
Ground state	Solution sought	Geometric fixed point ξ
Annealing	External (temperature)	Internal (K_{frak} correction)
Determinism	Hidden deterministic	Explicitly deterministic

Formal Correspondence

The complete mapping between T0 geometry and the Ising formalism reads:

$$\sigma_i \leftrightarrow \text{sign}(T(x, t)_i - \bar{T}) \quad [\text{spin from time field}] \quad (9)$$

$$J_{ij} \leftrightarrow \langle T(x, t)_i \cdot T(x, t)_j \rangle_\xi \quad [\text{coupling from } \xi \text{ geometry}] \quad (10)$$

$$h_i \leftrightarrow \frac{\partial \xi}{\partial T(x, t)_i} \quad [\text{external field from } \xi \text{ gradient}] \quad (11)$$

$$H_{\text{Ising}} \leftrightarrow E_{\text{T0}} = \int |\nabla T(x, t)|^2 d^4x \quad [\text{energy functionals identical}] \quad (12)$$

The minimum of the T0 energy functional (12) is identical to the ground state of the associated Ising Hamiltonian – with the crucial difference that in T0 all quantities (J_{ij} , h_i , H) emerge from a single parameter ξ .

The Central Consequence

Fundamental Implication of the T0 Ising Machine

If T0 is correct, then every physically realizable optimization problem is already encoded as a geometric structure in the mass field $m(x, t)$. A T0 Ising Machine would not optimize – it would *read* a physical reality. The solution is not the result of a search, but the unveiling of an already-existing geometric truth.

.7 Open Questions and Research Directions

Theoretical Openness

- What is the explicit geometric propagator $G(x_i, x_j)$ in the T0 mass field? What symmetries does the torus topology enforce?
- Is there a natural restriction on which NP-hard problems are representable through T0 geometry – or are all problems in principle encodable?
- How does K_{frak} behave as an annealing parameter for different problem classes? Are there universal scaling laws?
- What is the connection between the torus structure of the T0 universe and optimal CIM topologies (toroidal vs. Möbius-band couplings)?

Experimental Directions

- **Coupling structure comparison:** compare the coupling matrix of existing CIMs with T0 field propagator predictions. Are there systematic deviations pointing to geometric constraints?
- **Toroidal CIM topologies:** do toroidal CIM architectures show performance advantages over open graphs that go beyond combinatorial explanations?
- **T0 qubit test:** do the T0 qubits defined in (5) show characteristic signatures in existing photonic quantum computing experiments?
- **Testable prediction:** in a CIM with N spins, the T0 correction factor $\mathcal{D}(N) = \exp(-\xi \ln N / D_{\text{frak}})$ should produce a measurable $\ln(N)$ -scaling residual that is qualitatively distinct from standard error models ($1/N$ or e^{-N}).

Connection to Scramblon Physics

This document extends [10], which analyzed the T0 connection to scramblon physics. Combining both approaches yields a complete picture:

- **Scramblon theory:** dynamic description – how quantum information scrambles in a CIM-like system
- **T0 Ising Machine:** geometric description – which fundamental couplings follow from the spacetime structure
- **Synthesis:** scramblons describe the *dynamics* of information spreading; T0 describes the *geometry* of the coupling structure

.8 Summary

We have identified and systematically analyzed six structural parallels between Coherent Ising Machines and the T0 theory:

Table 8: Summary: Six parallels T0 ↔ CIM

#	Parallel	CIM	T0
1	Determinism	Hidden (OPO)	Explicit (geometry)
2	Ground state	Solution sought	Fixed point ξ
3	Qubit structure	Phase $0^\circ/180^\circ$	$T \cdot m = 1$ states
4	Couplings J_{ij}	Externally programmed	Geometrically emergent
5	Annealing	External temperature	K_{frak} correction
6	Topology	Arbitrary graph	4D torus

T0 does not merely provide a conceptual parallel to the CIM – it offers a fundamentally deeper justification: optimization is not a computation executed on an artificially constructed system, but physical relaxation into a geometric truth already inscribed in the mass field of the universe.

Central Result

The formal correspondence

$$H_{\text{Ising}} \leftrightarrow E_{\text{T0}} = \int |\nabla T(x, t)|^2 d^4x \quad (13)$$

shows: Ising optimization and T0 energy minimization are mathematically equivalent formulations of the same geometric principle – with the decisive difference that in T0 *all* couplings emerge parameter-free from $\xi = 4/30000$.

Bibliography

- [1] T. Inagaki et al., A coherent Ising machine for 2000-node optimization problems, *Science* **354**, 603 (2016).
- [2] P. L. McMahon et al., A fully programmable 100-spin coherent Ising machine with all-to-all connections, *Science* **354**, 614 (2016).
- [3] Y. Yamamoto et al., Coherent Ising machines – optical neural networks operating at the quantum limit, *npj Quantum Inf.* **3**, 49 (2017).
- [4] T. Honjo et al., 100,000-spin coherent Ising machine, *Sci. Adv.* **7**, eabh0952 (2021).
- [5] J. Pascher, T0 Theory: Time-Mass Duality as a Fundamental Principle, T0 Documentation Series, Document 003 (2025).
- [6] J. Pascher, The Geometric Origin Parameter $\xi = 4/30000$, T0 Documentation Series, Document 009 (2025).
- [7] J. Pascher, T0 Quantum Computing: Qubits and Logic Gates, T0 Documentation Series, Document 147 (2025).
- [8] J. Pascher, Fractal Corrections and the g-2 Anomaly, T0 Documentation Series, Document 018 (2025).
- [9] J. Pascher, The Universe as a 4D Torus Crystal: Geometric Cosmology, T0 Documentation Series, Document 026 (2025).
- [10] J. Pascher, T0 Projection onto Scramblon Physics, T0 Documentation Series, Document 148 (2026).